ASSIGNMENT 2

(SIR MODEL)

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For

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SIR MODEL EXERCISE

1. When a high γ or low β value is used, the trajectory does not return to the x-axis after peaking. Why might this be happening?

When β (infection rate) is high or γ (recovery rate) is low, a large part of the the population remains infected for a long time. The system stabilizes with a persistent level of infection instead of collapsing to the x-axis.

2. We have been initializing the models with 99.99% of the population as susceptible, and only 0.01% infected. How are the phase planes affected when changing this proportion?

As the number of infected persons in the population rises, the phase plane trajectory shifts upwards and leftwards (decreased susceptible population, increased infected population). The phase plane eventually peaks and then declines.

If the population gets infected slowly, the curve in the model shifts slowly and stays close to the susceptible axis for longer. This results in a gentle curve. However, if β is high, exponential growth sets in and the curve peaks faster, shifting the curve leftward. This results in a steep curve.

3. If we could reduce the infection transmission rate (e.g., through social distancing or vaccination), what do you think will happen to the nullclines? Then how would the epidemic curve change? What if this value increases?

A reduction in β (transmission rates) reduces the slope of the infection nullcline (indicating slower outbreaks).

The epidemic curve will have a lower peak and appears flatter (indicating less infections).

Increasing β will therefore have the opposite effect. First, a steeper slope on the infection nullcline (indicating faster outbreaks). Second, the epidemic curve has higher peaks (indicating more total infections).

4. If we could increase recovery rate γ, how would the curve change?

Assuming that in this model recovered individuals gain immunity from the disease, a higher recovery rate pulls people out of the infectious pool sooner and reduces transmission opportunities.

A higher γ gives an epidemic curve that is shorter and with a lower peak. Infections decline quickly after the peak.

5. If we reduced β while at the same time increasing γ how would that alter the shape of the epidemic curve?

Lower β reduces spread of the infection and a higher γ accelerates recovery from the infection. The epidemic curve becomes flatter, shorter and narrower with fewer people getting infected and a quicker resolution of the disease/epidemic. This combination is the most ideal resolution for any epidemic.

6. How can we estimate how long it will take for the epidemic to be at its peak and how long it will stay at its peak? This question is answered using phase planes and all clients in Emily's video.

Emily's video shows how trajectories bend toward the nullclines, giving a visual estimate of peak timing. A nullcline is simply the lines where the rate of change of a variable is zero.

The time for the epidemic to peak can be estimated by observing β (transmission rate), γ (recovery rate) and initial conditions. A higher β means the peak is reached quicker, and a higher γ means the duration at the peak is shorter.

7. How can the SIR model help governments decide when to lift restrictions?

Using the model can help predict when infections will decline below a safe threshold. This allows the government to assess how well interventions are working (e.g.,social distancing, vaccinations, treatment programs, etc). Restrictions can then be lifted when transmissions are reasonably low and recovery rates are high. Such decision making allows the government to ensure healthcare systems are not overwhelmed.

8. Can you now be able to answer the question you were asked at the start i.e can we predict when this outbreak will be at its peak, how many people will get infected and what interventions could stop it?

Yes, I can. I’d determine the peak of the outbreak by plotting the transmission rates against the recovery rates. Total infections would be given by the number of the susceptible population getting infected less/minus the number of the infected population recovering.

Interventions can be to lower transmission (e.g., social distancing, vaccination) or to raise recovery (e.g., treatment). I would know what interventions were working well by observing how the graph nullcline reacts and how the infected population changes after interventions were applied.